(iv) $2^0 = (1000)^0$

(v) $2^3 < 3^2$

3. Find the value of the following.

(i) 2^6 (ii) 11^2 (iii) 5^4 (iv) 9^3

4. Express the following in exponential form.

(i) $6\times6\times6\times6$ (ii) $t\times t$ (iii) $5\times5\times7\times7\times7$ (iv) $2\times2\times a\times a$

5. Express each of the following numbers using exponential form.

(i)512 (ii)343 (iii)729 (iv)3125

6. Identify the greater number in each of the following.

(i) 6^3 or 3^6 (ii) 5^3 or 3^5 (iii) 2^8 or 8^2

7. Simplify the following.

(i) $7^2\times3^4$ (ii) $3^2\times2^4$ (iii) $5^2\times10^4$

8. Find the value of the following.

(i) $(-4)^2$ (ii) $(-3)\times(-2)^3$ (iii) $(-2)^3\times(-10)^3$

9. Simplify using laws of exponents.

(i) $3^5\times3^8$ (ii) $a^4\times a^{10}$ (iii) $7^x\times7^2$ (iv) $2^5\div2^3$

(v) $18^8\div18^4$ (vi) $\left(6^4\right)^3$ (vii) $\left(x^m\right)^0$ (viii) $9^5\times3^5$

(ix) $3^y\times12^y$ (x) $25^6\times5^6$

10. If $a=3$ and $b=2$, then find the value of the following.

(i) a^b+b^a (ii) a^a-b^b (iii) $(a+b)^b$ (iv) $(a-b)^a$

11. Simplify and express each of the following in exponential form:

(i) $4^5\times4^2\times4^4$ (ii) $\left(3^2\times3^3\right)^7$ (iii) $\left(5^2\times5^8\right)\div5^5$

(iv) $2^0\times3^0\times4^0$ (v) $\dfrac{4^5\times a^8\times b^3}{4^3\times a^5\times b^2}$

Objective type questions

12. $a\times a\times a\times a\times a$ is equal to

(i) a^5 (ii) 5^a (iii) $5a$ (iv) $a+5$

13. The exponential form of 72 is

(i) 7^2 (ii) 2^7 (iii) $2^2\times3^3$ (iv) $2^3\times3^2$

14. The value of x in the equation $a^{13}=x^3\times a^{10}$ is

(i) a (ii) 13 (iii) 3 (iv) 10

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